



Python Programming - 2301CS404

Lab - 10

Roll No : 310

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01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string
- line by line
- in the form of a list

```
In [9]: fp = open("abc.txt", "r")
print(fp.read())
print(fp.readline())
fp.seek(0,0)
print(fp.readlines())
fp.close()
```

Hello World

Hi, how are you?

Good Morning

```
['Hello World\n', 'Hi, how are you?\n', 'Good Morning']
```

02) WAP to create file named "new.txt" only if it doesn't exist.

```
In [16]: fp = open("new.txt", "w")
fp.write("Hello World")
fp.close()

fp = open("new.txt", "r")
```

```
print(fp.read())
fp.close()
```

Hello World

03) WAP to read first 5 lines from the text file.

```
In [26]: with open("abc.txt", "r") as file:
          fdata = file.readlines()
          for i in fdata[:5]:
              print(i.strip())
```

Hello World
Hi, how are you?
Good Morning
This is python lab.
File/IO handling

04) WAP to find the longest word(s) in a file.

```
In [27]: def find_longest_words(filename):
          with open(filename, 'r') as file:
              words = file.read().split()
          if not words:
              return "The file is empty."
          max_length = len(max(words, key=len))
          longest_words = [word for word in words if len(word) == max_length]
          return list(set(longest_words))
          file_path = 'abc.txt'
          print("Longest word(s):", find_longest_words(file_path))
```

Longest word(s): ['handling']

05) WAP to count the no. of lines, words and characters in a given text file.

```
In [28]: lines=0;words=0;chars=0
          with open("new.txt", "r") as file:
              for line in file:
                  lines += 1
                  words += len(line.split())
                  chars += len(line)
          print(lines)
          print(words)
          print(chars)
```

7
8
45

06) WAP to copy the content of a file to the another file.

```
In [29]: fp = open("new.txt", "r")
          file = fp.read()
          newFp = open("copy.txt", "w")
          newFp.write(file)
          print("File Copied...")
```

```
fp.close()
newFp.close()
```

File Copied...

07) WAP to find the size of the text file.

```
In [35]: fp = open("copy.txt", "r")
print(fp.seek(0,2))
fp.close()
```

51

08) WAP to create an UDF named frequency to count occurrences of the specific word in a given text file.

```
In [37]: def frequency(word):
        fp = open("copy.txt", "r")
        words = fp.read().split()
        fp.close()

        freq = words.count(word)
        print(freq)
word = input("Enter the word to count freq:")
frequency(word)
fp.close()
```

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09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

```
In [44]: fp = open("marks.txt", "w")
max1 = 0
for i in range(1,6):
    mark = input("Enter marks:")
    fp.write(mark+"\n")
fp.close()
fp=open("marks.txt", "r")
m = fp.read().split()
print(m)
for i in m:
    i=int(i)
    if(int(i)>max1):
        max1=i
print(max1)
fp.close()
```

['1', '2', '3', '4', '5']

5

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

```
In [ ]: fp = open("primenumbers.txt", "w")
        for i in range(1,101):
            fp.write("", "r")
```

11) WAP to merge two files and write it in a new file.

```
In [1]: file1 = open("file1.txt", "r")
        file2 = open("file2.txt", "r")
        merged_file = open("merged.txt", "w")
        content1 = file1.read()
        content2 = file2.read()
        merged_file.write(content1)
        merged_file.write("\n")
        merged_file.write(content2)
        file1.close()
        file2.close()
        merged_file.close()
        print("Files merged successfully!")
```

Files merged successfully!

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

```
In [2]: word1 = "Python"
        word2 = "Java"
        with open("input.txt", "r") as file:
            content = file.read()
            updated_content = content.replace(word1, word2)
        with open("output.txt", "w") as file:
            file.write(updated_content)
        print("Word replaced successfully!")
```

Word replaced successfully!

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

```
In [7]: with open("copy.txt", "r") as file:
        print("Initial position:", file.tell())
        data = file.read(5)
        print("Read data:", data)
        print("Position after reading 5 chars:", file.tell())

        file.seek(0, 0)
        print("\nAfter seek(0,0):", file.tell())
        print("Read again:", file.read(5))
```

Initial position: 0
Read data: Hello
Position after reading 5 chars: 5

After seek(0,0): 0
Read again: Hello